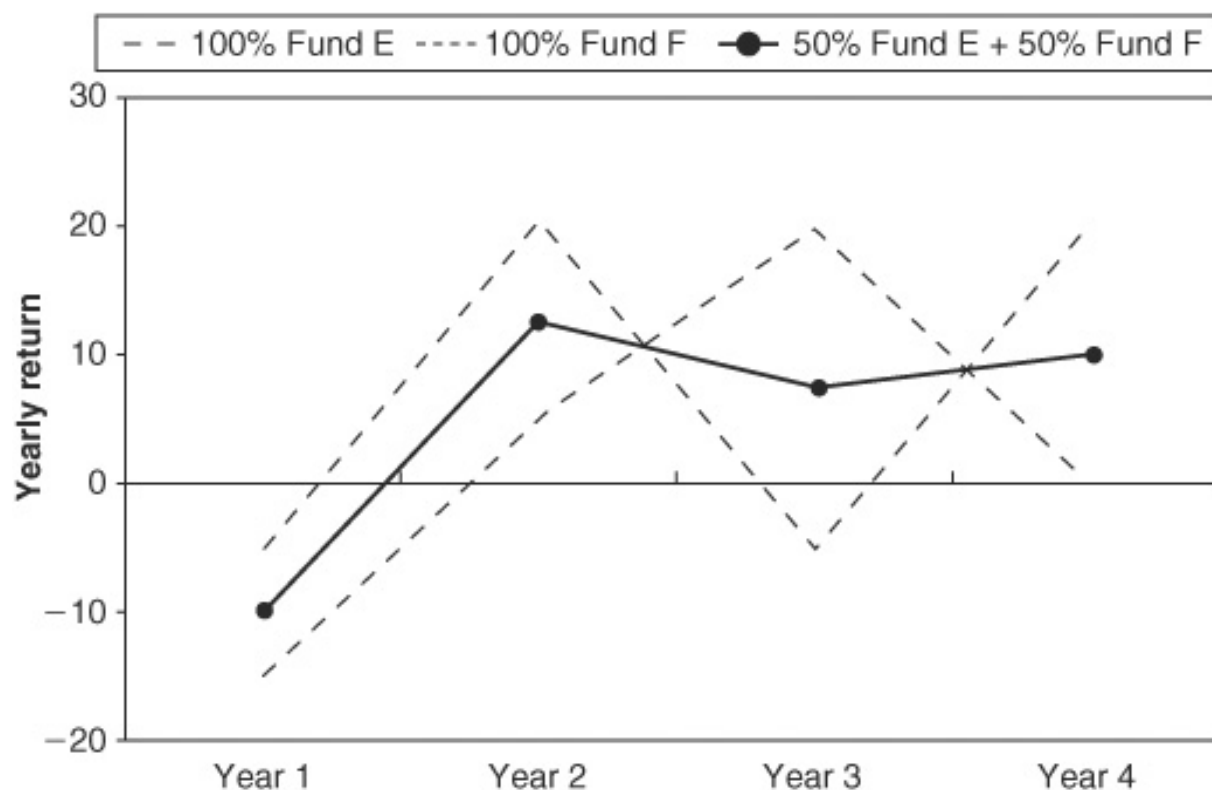


FIGURE 3-4

Noncorrelation Year-over-Year Returns

**TABLE 3-2**

Relationship between Correlation and Portfolio Return

Portfolio	Correlation of Assets	Simple Average Return	Compounded Return	Standard Deviation (σ)
1: 50% C + 50% D	-1.0	5.0%	5.0%	0%
2: 50% E + 50% F	0.0	5.0%	4.6%	10%
3: 50% A + 50% B	+1.0	5.0%	4.2%	14%

[Figure 3-5](#) illustrates the risk reduction benefit created by adding asset classes that have low or negative correlation with each other. Portfolio risk is reduced as correlation shifts from positive to negative, and the efficient frontier bows out further to the left, toward an area of lower portfolio volatility. This is the essence of risk reduction using asset allocation.